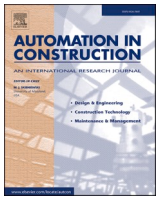
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Efficient edge-cloud digital twin for real-time SHM with server-deployed FEA and fast particle swarm optimization

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ABSTRACT

Digital twins (DTs) are increasingly applied throughout the life cycle of structures for health monitoring and decision-making. To meet real-time requirements, this paper presented an efficiency-enhanced DT framework based on an edge-cloud collaborative architecture and three innovations. First, a Python-based finite element (FE) analysis program with a precomputed stiffness matrix strategy significantly accelerated structural load identification and response computation. Second, a deviation severity coefficient with adaptive weighting effectively detected sensor faults and improved identification robustness. Third, a modified particle swarm optimization algorithm with population differentiation and local acceleration strategies further enhanced identification accuracy and efficiency. The system was validated in a bridge pier replacement project under five construction scenarios. The results demonstrated a sub-300 ms total response time, a 70-fold reduction in one-time FE computation, a 70 % decrease in optimization cost, stress prediction errors below 10%, and safety warnings for stress levels exceeding design values by over threefold.

1. Introduction

This section introduces the technical context required for understanding the proposed structural health monitoring (SHM) framework. It first discusses the fundamental principles and practical challenges of SHM, then outlines the motivations driving this study and summarizes its main contributions.

1.1. Background of structural health monitoring

Structural safety assurance necessitates a systematic evaluation of all foreseeable limit states and accidental risks during the design phase, encompassing the structure's entire service life [1,2]. This approach requires rigorous deterministic and probabilistic analyses of extreme loading scenarios, such as peak operational loads, foundation displacement thresholds, and accidental overload conditions [3–7]. Under these limit states, induced stresses and deformations must be maintained below both the ultimate material strength and permissible structural deformation limits to achieve quantified system reliability metrics. However, the following limitations significantly constrain the precision of existing methodologies. First, the structural system inherently diverges from its as-designed configuration due to material heterogeneity, coupled with performance variability, progressive degradation, and uncertainties emerging throughout construction and operational phases [8–11]. Applying idealized design models introduces inherent discrepancies that critically undermine structural integrity assessments. Second, boundary conditions such as soil-structure interactions resist precise numerical simulation due to their environmental sensitivity and temporal variability [12–14]. Consequently, the boundary conditions assumed during design-phase safety evaluations frequently diverge from field observations. Furthermore, critical scenarios often remain unaddressed in conventional design frameworks due to methodological limitations, economic constraints, or the probabilistic nature of unforeseen extreme events [15–17]. As a result, the deployment of SHM systems throughout the structural life cycle has become imperative to mitigate risks arising from indeterminate material properties, fluctuating boundary constraints, and stochastic loading scenarios [18,19]. Real-time and continuous monitoring of loading spectrum evolution and

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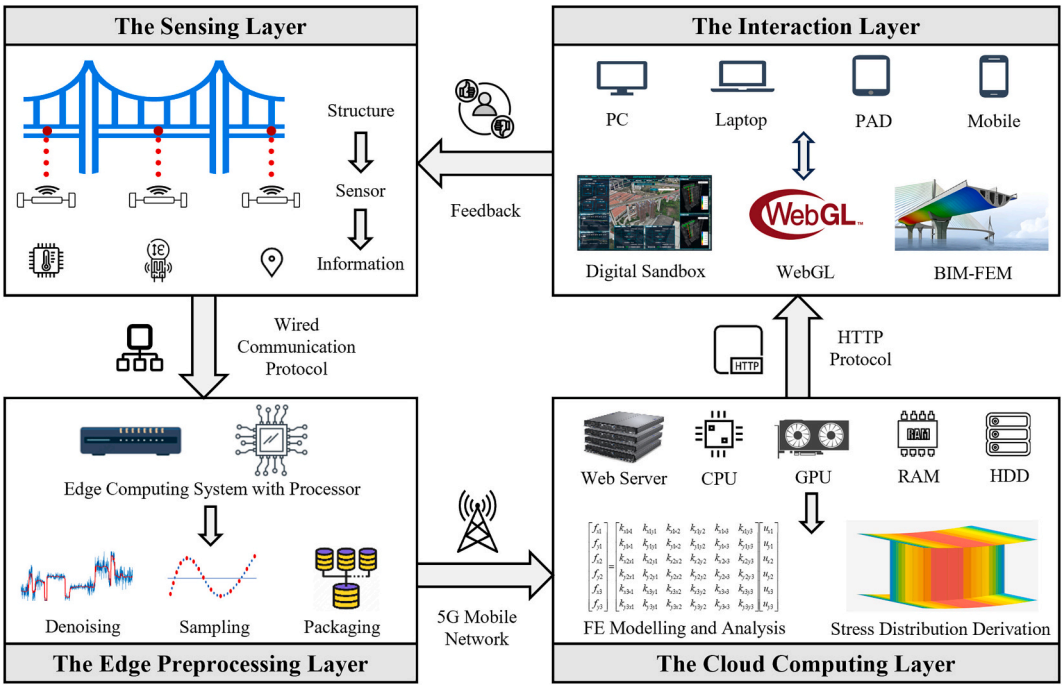


Fig. 1. Framework of digital twin for real-time structural health monitoring.

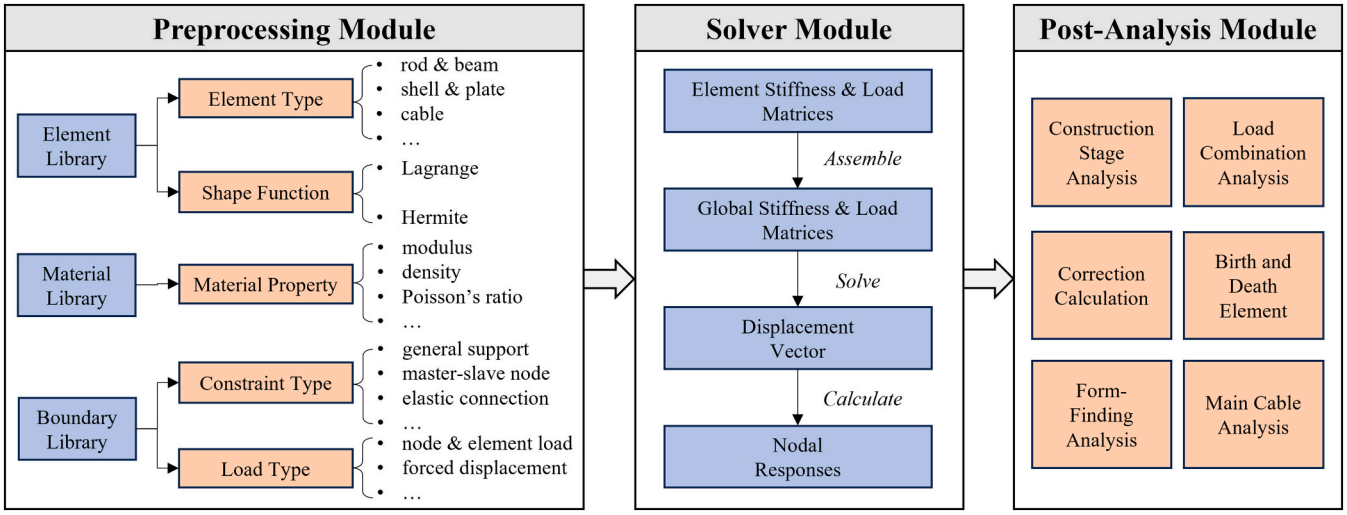


Fig. 2. Flow of finite element analysis.

associated structural responses enables reliable safety evaluations that accurately characterize in-service structural performance [20,21].

A SHM system typically comprises two core components: sensing networks and analytical diagnostics. It integrates sensor network measurements of structural-environmental parameters with data-informed prognostics to evaluate real-time integrity and forecast evolutionary performance states [22,23]. Currently, the sensing network prioritizes specialized sensors and advanced transmission techniques to enable real-time state acquisition. For example, Trapani et al. [22] implemented microcontroller-integrated piezoresistive ceramic sensors to directly quantify stress variations in masonry systems via electromechanical coupling, circumventing conventional strain-derived stress computation. As for data transmission, Ma et al. [24] implemented a long range radio (LoRa) wireless ad hoc network to transmit sensor data from monitoring nodes to local terminals, with General Packet Radio Service (GPRS) networks relaying the data to a centralized service platform. Hu et al. [25] utilized HTTP web protocols over wireless local area networks (WLANs) to enable communication between sensors and computer networks. Abner et al. [26] adopted a Bluetooth Low Energy (BLE) mesh network to facilitate multi-hop communication between sensor nodes, relay nodes, and station nodes, leveraging self-healing capabilities for dynamic path optimization. Mukherjee et al. [27] employed 5G networks to transmit sensor data from edge devices to remote cloud servers. Xu et al. [28] developed a dynamic wireless sensor network (DWSN) using tree-topology routing with address reallocation mechanisms to adapt to structural movements.

The analytical diagnostics emphasizes surrogate model development and deployment, typically leveraging finite element modeling (FEM) as the predominant paradigm to balance computational efficiency with high-fidelity structural representation [29,30]. Structural integrity assessment is performed through systematic comparison of field-measured structural responses against model-generated predictions,

Algorithm 1: Hybrid Load Identification and Stress Distribution Determination

Input: $\boldsymbol{\sigma}_{meas}$: a N x 1 vector containing measured stresses; $\boldsymbol{S}$: a N x M stress identification matrix; FEM: the structural FE model computing stress responses corresponding to the applied load; α_{th}: threshold of deviation severity

Output: $\boldsymbol{F}_{final}$: identified equivalent loads; $\boldsymbol{\sigma}_{global}$: determined global stress distribution

// **Phase 1: Initial load and deviation severity determination**

$\boldsymbol{F}_{all} \leftarrow \boldsymbol{S}^{+} \cdot \boldsymbol{\sigma}_{meas}$

$\boldsymbol{\sigma}_{calc} \leftarrow \mathrm{FEM}(\boldsymbol{F}_{all})$

Calculate deviation severity

for $i \leftarrow 1$ *to* N **do**

  $\boldsymbol{S}_{temp} \leftarrow \mathrm{RemoveRow}(\boldsymbol{S}, i)$ *# Remove the i-th row from stress identification matrix*

  $\boldsymbol{\sigma}_{meas_temp} \leftarrow \mathrm{RemoveRow}(\boldsymbol{\sigma}, i)$ *# Remove the i-th element from measured stress vector*

  $\boldsymbol{F}_{temp} \leftarrow \boldsymbol{S}_{temp}^{+} \cdot \boldsymbol{\sigma}_{meas_temp}$

  $\boldsymbol{\sigma}_{calc_temp} \leftarrow \mathrm{FEM}(\boldsymbol{F}_{temp})$

  $error_sum \leftarrow 0$

  for $j \leftarrow 1$ *to* N **do**

    if $j \neq i$ **then**

      $error_sum \leftarrow error_sum + (\boldsymbol{\sigma}_{calc_temp,j} - \boldsymbol{\sigma}_{meas,j})^2$

  $\alpha_i \leftarrow (\boldsymbol{\sigma}_{calc,j} - \boldsymbol{\sigma}_{calc_temp,j})^2 / error_sum$ *# Calculate deviation severity at the i-th sensor location according to Equation 4*

// **Phase 2: Load identification**

$\boldsymbol{W} \leftarrow \boldsymbol{W}_{init}$ *# Initialize all sensor weights to 1*

for $i \leftarrow 1$ *to* N **do**

  if $\alpha_i > \alpha_{th}$ **then**

    $\boldsymbol{W}_i \leftarrow \mathrm{Lower}(\boldsymbol{W}_i)$ *# Lower weights of sensor locations with high deviation severity values*

$f(\boldsymbol{F}) \leftarrow \sum_{i=1}^{N} W_i \cdot (\boldsymbol{\sigma}_{calc,i}(\boldsymbol{F}) - \boldsymbol{\sigma}_{meas,i})^2$ *# Define the objective function with the differences between measured and FEM-predicted stresses*

$\boldsymbol{F}_{\mathrm{final}} \leftarrow \arg\min_{\boldsymbol{F}} f(\boldsymbol{F})$

// **Phase 3: Stress distribution determination**

$\boldsymbol{\sigma}_{global} \leftarrow \mathrm{FEM}(\boldsymbol{F}_{final})$

return $\boldsymbol{F}_{final}, \boldsymbol{\sigma}_{global}$

where the model operates under parameterized representations of undamaged structural properties, in-situ loading profiles, and ambient environmental conditions monitored in real-time. However, the convergence of three critical developments, namely real-time operational demands in SHM, multi-module integration requirements in digital twin (DT) frameworks [31], and enhanced machine learning (ML) capabilities in system relationship extraction, has driven a paradigm shift toward data-driven surrogate modeling architectures. Training data can be derived from physics-based simulations, in-situ measurements, and synthetic data augmentation, with diverse modalities including cross-sectional, sequential, and image-based formats [32–36].

1.2. Motivation

As outlined above, SHM has evolved into a multidisciplinary field integrating hardware innovation with advanced system analytics. These advancements have progressively addressed the critical demands for real-time performance and accuracy in SHM systems. However, existing methodologies still face persistent limitations that arise from the inherent trade-off between accuracy and computational efficiency. As a result, monitoring performance remains suboptimal, underscoring research gaps that require systematic investigation and methodological innovation.

Dense sensor arrays with high sampling rates are always employed to achieve comprehensive structural characterization [21]. Bridge health monitoring systems, for instance, can routinely generate multi-gigabyte daily datasets through continuous sensing modalities, creating substantial challenges in data transmission bandwidth and computational resource allocation [37]. Although high-bandwidth communication channels are theoretically essential for maintaining data fidelity and supporting real-time analysis [38], practical deployment is often constrained by harsh environmental conditions and hardware limitations, which make such infrastructure difficult to implement in the field [39]. Furthermore, raw sensor outputs predominantly contain extraneous information unrelated to structural integrity assessment, necessitating advanced signal processing techniques (e.g., Fourier spectral analysis, wavelet transform, time-domain feature extraction) to distill physically meaningful indicators from the acquired datasets [25,37,40].

Data-driven surrogate models, particularly ML approaches, have gained traction as computationally efficient alternatives to FEM for real-time SHM in large-scale systems. Nevertheless, their practical implementation is constrained by three fundamental limitations: the sampling bias inherent in training datasets, the architectural constraints of surrogate models, and the absence of mathematically rigorous frameworks

Table 1
Fundamental mechanism of Fast-PSO.

Equation	Action	Mathematical Form	Term Explanation
5	Velocity update in standard PSO	$v_i(t+1) = wv_i(t) + c_1 r_1(\text{pbest}_i - x_i(t)) + c_2 r_2(x_{gbest} - x_i(t))$	$\mathbf{v}_i$ and $\mathbf{x}_i$: velocity and position of the i-th particle; t: iteration index; w: inertial weight; c_1 and c_2: cognition and social coefficients; r: random numbers in the interval [0,1]; $\mathbf{pbest}_i$ and $\mathbf{x}_{gbest}$: the historical best position of the i-th particle and the swarm.
6	Location update in standard PSO	$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1)$	
7	Exploration Dynamism calculation in Fast-PSO	$\alpha(\mathbf{x}_i) = \|\mathbf{x}_i - \mathbf{pbest}_i\|_2$	α: Exploration Dynamism Index.
8	Velocity update of Exploratory particles in Fast-PSO	$\mathbf{v}_i(t+1) = w_{exp}\mathbf{v}_i(t) + c_{1,exp} r_1(\mathbf{pbest}_i - \mathbf{x}_i(t)) + c_{2,exp} r_2(\mathbf{x}_{gbest} + \mathbf{L}(t) - \mathbf{x}_i(t))$	w_{exp}, $c_{1,exp}$, and $c_{2,exp}$: inertial weight, cognition coefficient, and social coefficient of Exploratory particles; $\mathbf{L}(t)$: random perturbation vector generated by the Lévy flight distribution.
9	Velocity update of Exploitative particles in Fast-PSO	$\mathbf{v}_i(t+1) = w_{expl}\mathbf{v}_i(t) + c_{1,expl} r_1(\mathbf{pbest}_i - \mathbf{x}_i(t)) + c_{2,expl} r_2(\mathbf{x}^*_{local} - \mathbf{x}_i(t))$	w_{expl}, $c_{1,expl}$, and $c_{2,expl}$: inertial weight, cognition coefficient, and social coefficient of Exploitative particles; $\mathbf{x}^*_{local}$: the optimal solution of the HPSS surrogate model.
10	Reinvigoration of Stagnant particles in Fast-PSO	$\mathbf{x}_i = r_3\mathbf{x}_{gbest} + (1-r_3)\overline{\mathbf{x}}_{exp} + \mathbf{L}(t)$	$\overline{\mathbf{x}}_{exp}$: the mean position of all Exploratory particles.
11	Particle group allocation in Fast-PSO	$\boldsymbol{\rho}(t+1) = g\left(\boldsymbol{\rho}(t), \{\mathbf{x}_{gbest}(k)\}_{k=t}^{k=t-4}\right)$	$\boldsymbol{\rho}$: group proportion vector; g: group allocation strategy.
12	Surrogate model of HPSS in Fast-PSO	$\widehat{f}(\mathbf{x}) = \sum_{j=0}^{P} \beta_j \phi_j(\mathbf{x})$	β and ϕ: coefficients and basis functions of the HPSS surrogate model;
13	Surrogate model training in Fast-PSO	$\boldsymbol{\beta} = \text{argmin}_{\boldsymbol{\beta}} \sum_{(\mathbf{x}_k, y_k) \in \mathbf{S}_{train}} [\widehat{f}(\mathbf{x}_k) - f(\mathbf{x}_k)]^2$	$f(\mathbf{x}_k)$ and $\widehat{f}(\mathbf{x}_k)$: the actual value and surrogate model prediction at position $\mathbf{x}_k$; $\mathbf{S}_{train}$: the training dataset of the HPSS surrogate model.

for quantifying error propagation and generalization boundaries. Current implementations demonstrate acceptable accuracy within their trained parameter spaces but exhibit significant performance deterioration under out-of-distribution (OOD) scenarios, particularly when confronted with uncharacterized loading patterns or emerging structural degradation modes [33]. Furthermore, architectural mismatch and data-model imbalance, which manifest as overfitting and representation gaps, remain poorly quantified when relying on conventional training-accuracy metrics [41,42]. Notably, the severe scarcity of anomaly data relative to normal operational measurements introduces intrinsic sampling skewness, while the topological complexity of engineered systems compounds diagnostic ambiguity, frequently preventing precise subsystem-level anomaly attribution. Although domain-informed ML approaches (e.g., physics-informed neural networks, PINN) show promise in enhancing model generalization and robustness [43,44], FEM retains superior predictive fidelity and reliability for safety-critical engineering applications [45,46].

1.3. Contributions and outline

To address persistent challenges in reconciling computational efficiency with diagnostic accuracy for real-time SHM, this study developed a DT framework integrating the following three core innovations:

- **Real-time FE execution**: A Python-based FE program was developed and implemented to execute structural response analysis through cloud computing. Computational efficiency was enhanced through precomputed stiffness matrix and sparse matrix protocols, satisfying sub-second latency constraints for real-time safety assessments.
- **Robust sensor fault detection**: A deviation severity coefficient was formulated to detect abnormal sensors. Adaptive weighting was applied to balance heterogeneous sensor data, thereby improving the robustness of system identification.
- **Efficient system identification**: A modified particle swarm optimization variant was designed to expedite system parameter identification, yielding times of convergence improvements without compromising precision for time-constrained diagnostics.

Although advanced hardware technologies such as edge-cloud integration, cloud computing, and 5G wireless networks are still employed in the proposed DT system for rapid data transmission and processing, the three innovations presented in this paper primarily target methodological advances in generic system simulation and identification. These contributions aim to provide adaptable improvements for future health monitoring tasks across different structural types, varying levels of complexity, and diverse hardware conditions. In addition to the fundamental concepts and recent advancements in SHM and DT presented in Section 1, the remainder of the paper is structured as follows: Section 2 delineates the constituent modules of the DT framework while elucidating technical implementations of three core contributions; Section 3 demonstrates the implementation process and device configurations of the proposed system through an engineering application; Section 4 conducts a comprehensive analysis of experimental findings, employing systematic comparison with conventional approaches to validate the enhanced performance of the developed strategies; and Section 5 synthesizes principal conclusions and outlook for future research.

2. Methodology

This section systematically describes the functional architecture of the proposed DT framework, summarizing the roles, data flow, and interactions among its core modules. These technical elements serve as the basis for the three methodological contributions that are elaborated in the subsequent three subsections.

2.1. Proposed framework of digital twin

The proposed framework, as depicted in Fig. 1, comprises three hierarchical hardware architecture layers: the sensing layer, the edge preprocessing layer, and the cloud computing layer. The sensing layer integrates a network of sensors to monitor external environmental parameters (e.g., temperature) and internal structural responses (e.g., strain), transmitting raw data to edge devices. The edge preprocessing layer executes localized lightweight computations, including feature extraction, data fusion, and forwarding operations, all tailored to task-specific requirements and data attributes. This step minimizes data volume and noise to be transmitted, optimizing transmission efficiency and quality. The cloud computing layer, serving as the framework's core computational component, integrates modules for data storage, caching, processing, and analytics to facilitate advanced simulation, analysis, and visualization. For example, in the engineering application introduced in Section 3, the cloud server performs FE analysis, dynamically updating structural loads and global stress distributions using real-time data to

Algorithm 2: Fast-PSO

Input: $f(\cdot)$: objective function; E_{max}: allowable evaluation number; T_{init}: initial iteration number of standard PSO; $\boldsymbol{\rho}_{init}$: initial proportion vector of Exploratory, Exploitative, and Stagnant particles; $T_{validate}$: number of cycles to validate historical best positions of Exploitative particles

Output: $\boldsymbol{x}_{gbest}$: global optimum solution; f_{gbest}: global best fitness

*// **Phase 1: Initialization***

Initialize $\boldsymbol{x}$ with Latin hypercube sampling; $\boldsymbol{\rho} \leftarrow \boldsymbol{\rho}_{init}$; $\{\boldsymbol{v}_i\}_{i=1}^{N} \leftarrow \mathbf{0}$; $\boldsymbol{pbest}_i \leftarrow \boldsymbol{x}_i$; $\{\boldsymbol{f}_{\mathrm{pbest},i}\}_{i=1}^{N} \leftarrow \infty$; $f_{gbest} \leftarrow \infty$; $\boldsymbol{x}_{gbest} \leftarrow \emptyset$

*// **Phase 2: Standard PSO***

for *iteration* $\leftarrow 1$ *to* T_{init} **do**
  # Evaluate all individuals and record best positions
  forall $i \in \{1, \ldots, N\}$ **do**
    $\boldsymbol{f}_i \leftarrow f(\boldsymbol{x}_i)$
    if $\boldsymbol{f}_i < f_{pbest,i}$ **then**
      $f_{pbest,i} \leftarrow \boldsymbol{f}_i$; $\boldsymbol{pbest}_i \leftarrow \boldsymbol{x}_i$
    if $\boldsymbol{f}_i < f_{gbest}$ **then**
      $f_{gbest} \leftarrow \boldsymbol{f}_i$; $x_{gbest} \leftarrow \boldsymbol{x}_i$
  forall $i \in \{1, \ldots, N\}$ **do**
    # Update all particle positions and velocities
    Update $\{\boldsymbol{v}_i\}_{i=1}^{N}$ and $\{\boldsymbol{x}_i\}_{i=1}^{N}$ according to Equation 5 and Equation 6

*// **Phase 3: Population differentiation***

while *iteration* $< E_{max}$ **do**
  Calculate Exploration Dynamism of all particles according to Equation 7
  Update $\boldsymbol{\rho}$ according to Equation 11
  forall $i \in \mathcal{C}_{exp}$ **do**
    # Process Exploratory particles
    $\boldsymbol{f}_i \leftarrow f(\boldsymbol{x}_i)$
    Update $\boldsymbol{f}_{\mathrm{pbest},i}$, $\boldsymbol{pbest}_i$, f_{gbest}, and x_{gbest}
    Update $\boldsymbol{v}_i$ and $\boldsymbol{x}_i$ according to Equation 8 and Equation 6
  forall $i \in \mathcal{C}_{expl}$ **do**
    # Process Exploitative particles
    Build surrogate model $\hat{f}(\boldsymbol{x})$ with $\boldsymbol{x}_{i \in \mathcal{C}_{\mathrm{expl}}}$ according to Equation 12 and Equation 13
    Obtain optimal solution of $\hat{f}(\boldsymbol{x})$ with mathematical optimization techniques
    Update $\boldsymbol{f}_{\mathrm{pbest},i}$ and $\boldsymbol{pbest}_i$
    Update $\boldsymbol{v}_i$ and $\boldsymbol{x}_i$ according to Equation 9 and Equation 6
  forall $i \in \mathcal{C}_{stag}$ **do**
    # Process Stagnant particles
    $\boldsymbol{v}_i \leftarrow 0$ and update $\boldsymbol{x}_i$ according to Equation 10
  if *iteration* $\equiv 0 \pmod{T_{validate}}$ **then**
    Validate $\boldsymbol{pbest}_i$ of Exploitative particles predicted by the surrogate model with actual values

return $\boldsymbol{x}_{gbest}, f_{gbest}$

ensure precise structural safety assessment. As an essential component of industrialized tools, the interaction layer is mainly responsible for information visualization of the DT model and subsequent decision making. It leverages WebGL technology for cross-platform accessibility, allowing structural condition monitoring through terminal devices such as PCs, laptops, tablets, and smartphones.

Data transmission between the sensing layer and the edge layer can utilize either wired or wireless communication protocols (e.g., Wi-Fi, Zigbee, LoRa), with selection criteria determined by specific engineering characteristics and operational requirements. Wi-Fi demonstrates superior transmission quality and bandwidth efficiency, though significant signal attenuation occurs in obstructed environments, making it suitable for applications with simple structural configurations requiring unobstructed sensor-terminal pathways. While Zigbee exhibits inferior

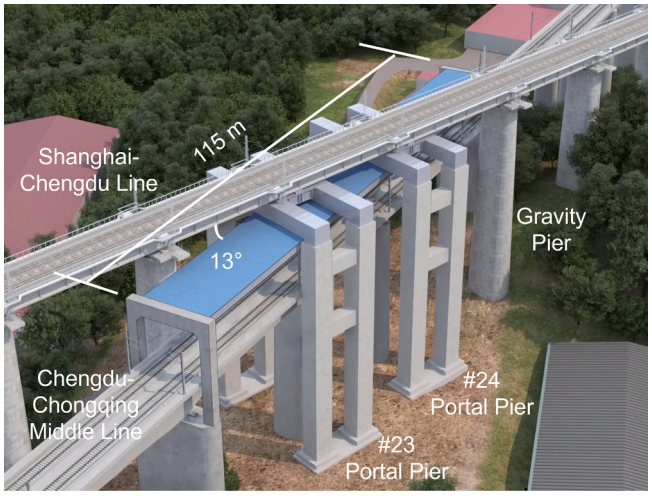


Fig. 3. Conceptual site overview diagram of the application.

Table 2
Information of investigated scenarios.

Scenario	Specific Task	Subject Component	Applied Force	Primary State
1	Constructing Platform	Trestle No.1	Self-weight of platform and workers	Eccentric compression of the trestle
2	Installing Steel Crossbeam	Trestle No.1	Self-weight of the steel crossbeam	Compression of the trestle
3	Lifting Main Girder and Cushion Beam	Trestle No.1	Self-weight of the main girder	Deformation of steel crossbeam and compression of the trestle
4	Existing Pier Cap Sliding	Trestle No.2	Existing pier cap sliding onto the trestle	Compression and eccentric tension of the trestle
5	Existing Pier Cap Lowering	Trestle No.2	Self-weight of the pier cap	Compression of the trestle

transmission quality and speed compared to Wi-Fi, its ultra-low power consumption renders it appropriate for long-term monitoring systems. In the engineering application, given the infrastructure's high-risk nature and stringent requirements for real-time data integrity, wired transmission was implemented to ensure optimal transmission efficiency and data fidelity.

Data transmission from the edge layer to the cloud can be achieved via wired (e.g., Ethernet, fiber optics) or wireless (e.g., 4G, 5G) methods. Wired systems ensure secure, high-fidelity data transmission during high-volume transfers, whereas wireless systems eliminate geographical constraints between endpoints. Given the server-edge distance in the engineering application, 5G technology was selected for its balanced consideration of transmission speed, quality, and cost efficiency. In contrast to MQTT, which prioritizes low power consumption over transmission stability and data integrity, the TCP/IP protocol was adopted for edge-to-cloud communication because its standardized architecture ensures robust reliability and industrial applicability. Furthermore, encryption and compression techniques were integrated to enhance security while preserving bandwidth efficiency.

The cloud computing layer employs distributed computing frameworks including Hadoop and Spark. Spark was specifically chosen in the engineering application for its Python-compatible API architecture and accelerated computational capabilities. It specifically aligns with two critical requirements: compatibility with existing Python-based FE analysis modules, and real-time processing capability essential for SHM. Processed data was subsequently transmitted via HTTP/HTTPS protocols to client terminals, enabling real-time visualization of analytical results and interactive models through standard web browsers.

2.2. *Python-based finite element program with cloud computing*

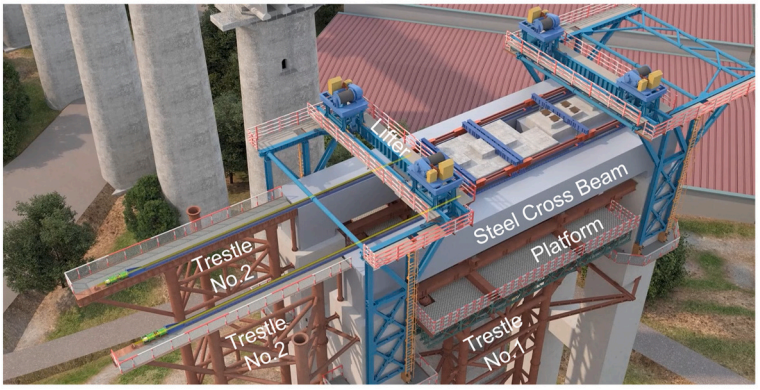
Conventional structural FE analysis predominantly relies on commercial desktop-based softwares (e.g., ABAQUS, ANSYS, Midas Civil, etc.), necessitating manual or scripted data transfer for sensing inputs and analytical outputs. This workflow fragmentation introduces computational bottlenecks that critically compromise structural monitoring system automation and efficiency. While commercial platforms emphasize visualization capabilities, especially in modeling and analysis as the key distinguishing feature, such functionality becomes non-essential in well-constrained application scenarios (e.g., specific load-response solutions at predefined critical nodes). Furthermore, graphical processing demands substantial GPU resources for rendering and computation, making visualization impractical for typical resource-constrained monitoring system architectures with limited computational and storage capacity. As illustrated in Fig. 2, the developed FE program only preserves the formulation, assembly, and computation of stiffness and force matrices [47]. To overcome client-side constraints in data and workflow management, the Python-implemented system employs MySQL for database operations, enabling full cloud-based functionality.

To ensure modeling efficiency for both engineering application and subsequent projects, the preprocessing module incorporates the predefined Element Library with common element types and shape functions, the Material Library containing elastic moduli, densities, and Poisson's ratios for diverse concrete and steel grades, and the Boundary Library defining various constraint and load conditions. The Python-based solver module enables flexible equation system resolution based on structural characteristics. For sparse global stiffness matrices, the SciPy sparse solver ("scipy.sparse.linalg.spsolve") is employed, while models with limited degrees of freedom utilize the direct solver from NumPy ("numpy.linalg.solve") to optimize computational efficiency. In the post-analysis module, Python APIs extract nodal and elemental displacement and stress responses, enabling specialized analyses such as staged construction simulation, load combination evaluation, and structural configuration optimization. Python's interoperability facilitates seamless integration of analysis results with other digital systems, including SHM platforms and engineering information management frameworks. This flexibility supports cross-platform data sharing, secondary development, and collaborative workflows.

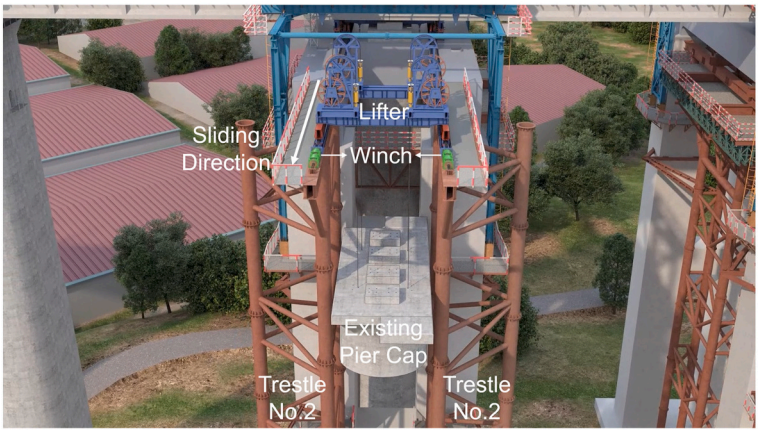
In addition to leveraging Python to achieve intensive FE analysis capabilities, the program employs two strategies to optimize SHM efficiency. The first is a computation acceleration strategy based on the fixed stiffness matrix. In structural analysis, when the topology and geometric configuration of the structure remain unchanged, the global stiffness matrix theoretically remains constant across varying external loading conditions. By reusing the precomputed global stiffness matrix for multi-load case analyses, the program minimizes redundant element-level stiffness calculations and global matrix assembly processes, substantially accelerating real-time structural response evaluation. The second approach focuses on sparse matrix storage and in-memory computing. Given the high sparsity of the FE-derived global stiffness matrices in the engineering application, the program adopts the Coordinate (COO) format to reduce storage requirements. Optimized data structures and storage schemes enable core computations, particularly the solution of large-scale linear systems, to be performed primarily in memory. This memory-centric strategy minimizes interactions with the hard disk, limiting them to necessary data output at the end of the analysis, thereby further reducing computational cycles and enhancing overall efficiency.



(a) Construction Site and Equipment Layout Photo



(b) Conceptual Diagram of View 1



(c) Conceptual Diagram of View 2

Fig. 4. Illustration of investigated scenarios.

2.3. Hybrid load identification with adaptive sensor weighting

During complex construction phases or under varying loading conditions, imposed structural loads are often uncertain or difficult to quantify precisely. This uncertainty poses a significant challenge for FE analysis, which requires explicit load inputs to compute structural responses. This study proposed a load identification method that determines the equivalent loads acting on a structure using monitored data, specifically stress measurements obtained from a limited number of instrumented locations. Subsequently, these inverted loads serve as inputs to the system for performing a global stress analysis. This enables the determination of stress distributions across all structural

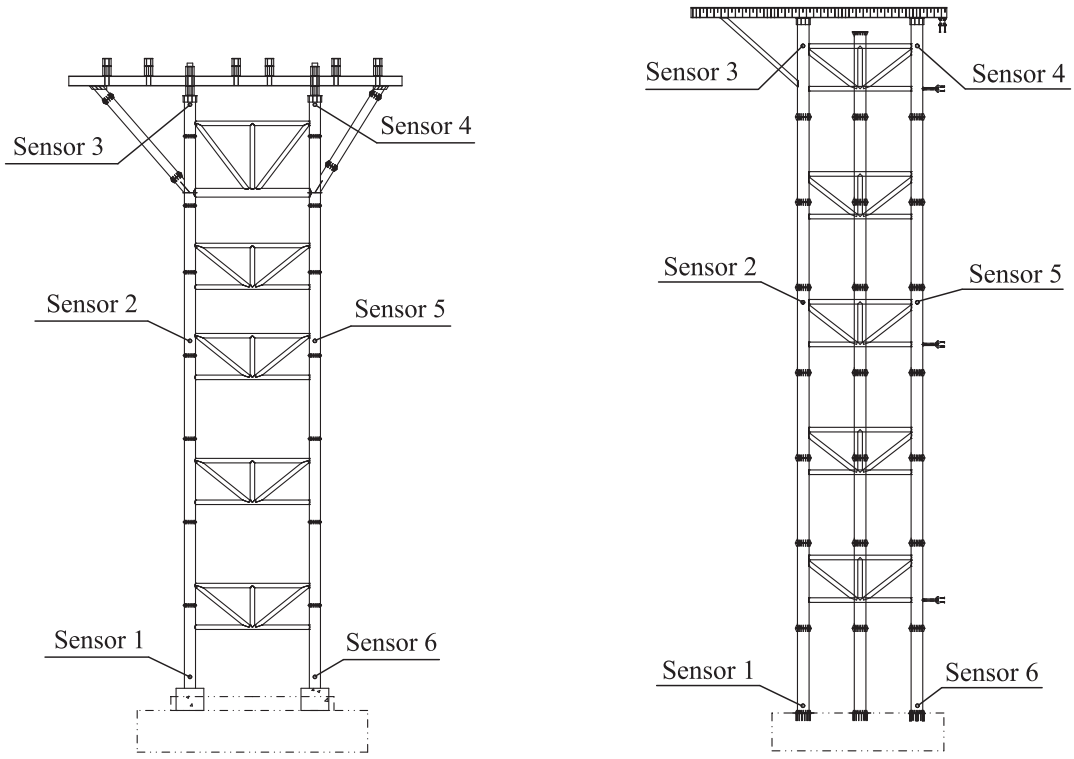


Fig. 5. Schematic diagram of sensor arrangement on trestles.

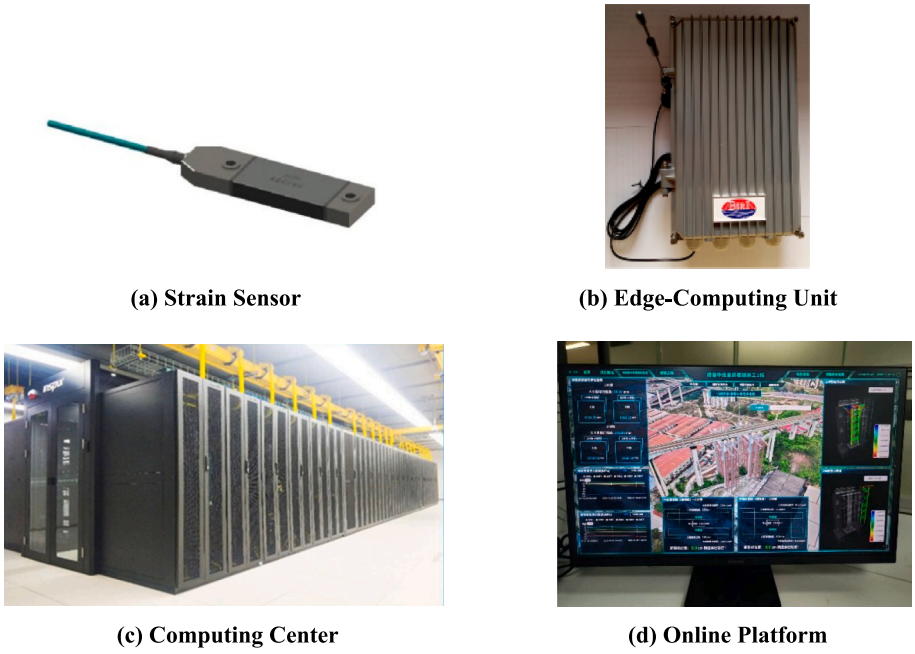


Fig. 6. Representative components of the digital twin system.

components, including potentially uninstrumented regions, thereby facilitating more comprehensive and timely structural safety monitoring.

As outlined in Algorithm 1, the initial step quantifies the relationship between structural loads and stresses. Under the assumptions of linear elasticity and small deformations, it is governed by a linear system as Eq. 1 [48],

$$\boldsymbol{\sigma} = \mathbf{SF} \tag{1}$$

where the stress influence matrix **S** is explicitly defined through the elasticity matrix, strain-displacement matrix, displacement transformation matrix, and the inverse of stiffness matrix [49]. In addition to matrix combination with theoretical structural configuration, the stress

Table 3
Device configurations of the digital twin system.

Layer	Component	Parameters
Sensing Layer	Strain Sensors	Measurement Range: ±3000 με Nonlinearity: < ±2 % FS Sensitivity: 500 με/mV/V Bridge Voltage: < 10 VDC
Edge Preprocessing Layer	Edge Computing Unit	Processor: digital signal processor (DSP) Number of Channels: 8 Power Consumption: 15 W
Cloud Computing Layer	Computing Center	CPU: 128 cores Memory (RAM): 64 GB GPU Memory: 24 GB
Interaction Layer	Online Platform	Visualization Engine: Three.js, Cesium Architecture: microservice architecture

influence matrix can be obtained from FE simulations as in this study using its concept that the component S_{ij} refers to the stress response at the i-th location with unit force acting on the j-th location. However, the stress influence matrix is frequently ill-conditioned or singular due to intrinsic limitations such as sparse sensor placement, measurement noise, and load distribution ambiguities [50]. This pathological conditioning renders the load inversion problem ill-posed, manifesting as solution non-uniqueness and extreme sensitivity to data perturbations. For singularity, the Moore-Penrose pseudoinverse of the stress influence matrix $\mathbf{S}^{+}$ can be utilized in the singular value decomposition (SVD) method as Eq. 2 [51],

$$\mathbf{S}^{+} = V\Sigma^{+}U^{T} \tag{2}$$

in which,

$$\mathbf{S} = U\Sigma V^{T} \tag{3}$$

where U and V are two orthogonal matrices; and Σ^{+} is the pseudoinverse of a diagonal matrix Σ.

For measurement noise, a leave-one-out (LOO) scheme was proposed in this study to assess sensor reliability and adjust sensor weight for load inversion. Inspired by techniques in sensor fault detection and isolation (FDI) [52] and outlier detection in structural identification [53], each sensor data is iteratively eliminated and its faulty potential is quantified by the "deviation severity (DS)" defined as Eq. 4,

$$\alpha_i = \frac{\left(\widehat{\sigma}_i - \widehat{\sigma}_i^{i}\right)^2}{\sum_{j=1}^{N-1}\left(\widehat{\sigma}_j^{i} - \sigma_j\right)^2} \tag{4}$$

where α_i is the deviation severity at the i-th sensor location; $\widehat{\sigma}_i$ is the predicted response at the i-th sensor location using all sensor data via load inversion; $\widehat{\sigma}_i^{i}$ and $\widehat{\sigma}_j^{i}$ are the predicted responses at the i-th and j-th sensor locations, respectively, with the i-th sensor data excluded from load inversion; σ_j is the measured response at the j-th sensor location; and N is the total number of sensor locations. Sensors with high DS values are assigned reduced weights in constructing the objective function to be minimized. Finally, loads that generated FE stresses consistent with measured values can be identified and used to determine

Table 4
Execution time (seconds) in digital twin workflow.

Scenario	Process				Total
	Data from Sensor Layer to Cloud Computing Layer	Load Identification	Stress Distribution Determination	Digital Twin Visualization	
1		0.206	0.0305		0.261
2		0.146	0.0301		0.204
3	N/A*	0.195	0.0306	N/A*	0.254
4		0.118	0.0252		0.177
5		0.107	0.0249		0.164

* The time costs associated with hardware and network configurations are included in the "Total" time shown in this table but are not reported separately.

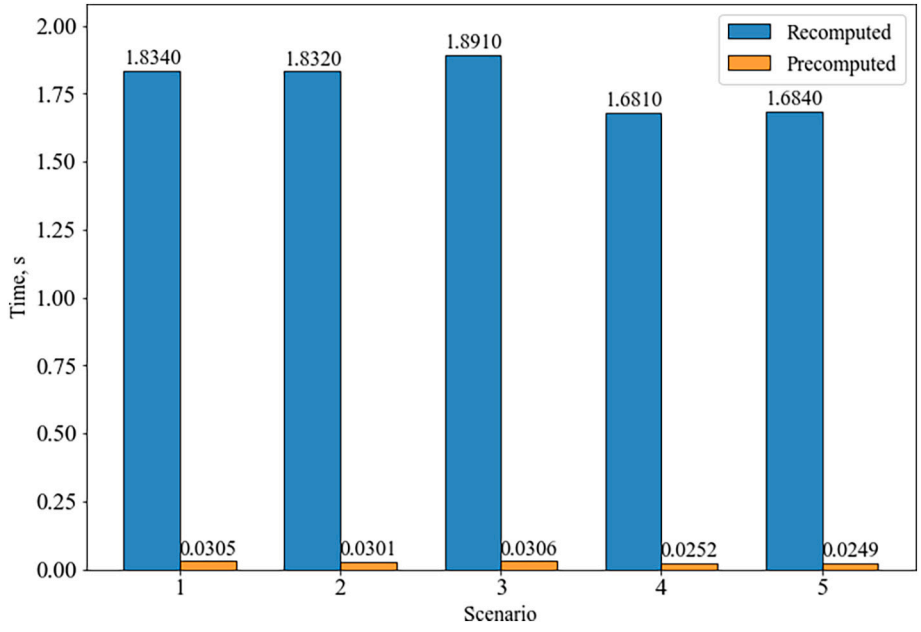


Fig. 7. Effect of stiffness matrix generation method on execution time (seconds) of stress distribution determination.

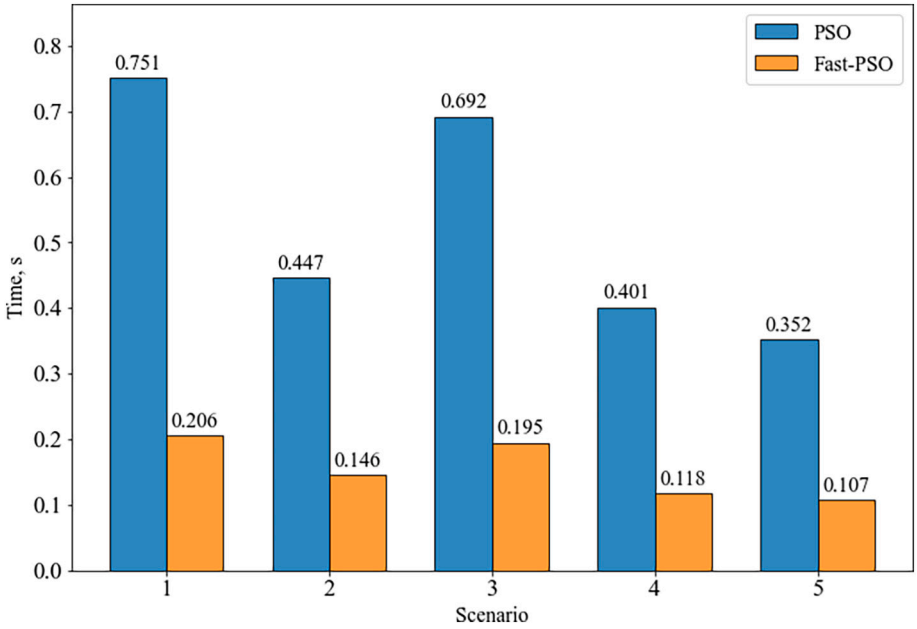


Fig. 8. Effect of optimization method on execution time (seconds) of load identification.

Table 5
Hyperparameters of PSO and Fast-PSO.

Hyperparameter	Range	Value	
		PSO	Fast-PSO
Population Size	–	20 for the engineering application and 50 for the benchmark functions	
Initial Proportion Vector of Exploratory, Exploitative, and Stagnation Particles	–	–	[0.4, 0.4, 0.2]
Surrogate Model of HPSS $\widehat{f}(\mathbf{x})$	–	–	Quadratic response surface model
Weight Coefficient, w (w_{exp} and w_{expl})	[0.4, 0.9]	0.729	
Cognition Coefficients, c_1 ($c_{1,exp}$ and $c_{1,expl}$)	[0, 4]	1.494	
Social Coefficients, c_2 ($c_{2,exp}$ and $c_{2,expl}$)	[0, 4]	1.494	

the structural stress distribution.

2.4. Fast-PSO algorithm

As established in the previous section, the load inversion technique which relies on matching FE-predicted responses with field measurements (i.e., Line 17 in Algorithm 1), constitutes a conventional FE model-updating problem. In this process, swarm intelligence algorithms are typically employed (i.e., Line 18 in Algorithm 1) to perform automated optimization [54,55]. Particle swarm optimization (PSO) is a classic swarm intelligence algorithm optimizing solutions through iterative updates of individual positions (i.e., solutions), and these updates are guided by the historical optimal positions of both the swarm and individual particles [56]. During this process, the algorithm requires iterative evaluations of the objective function to assess the positional quality (i.e., fitness) of particles. However, the computational complexity of massive structural FE models and data transfer delays across modules significantly reduce optimization efficiency through

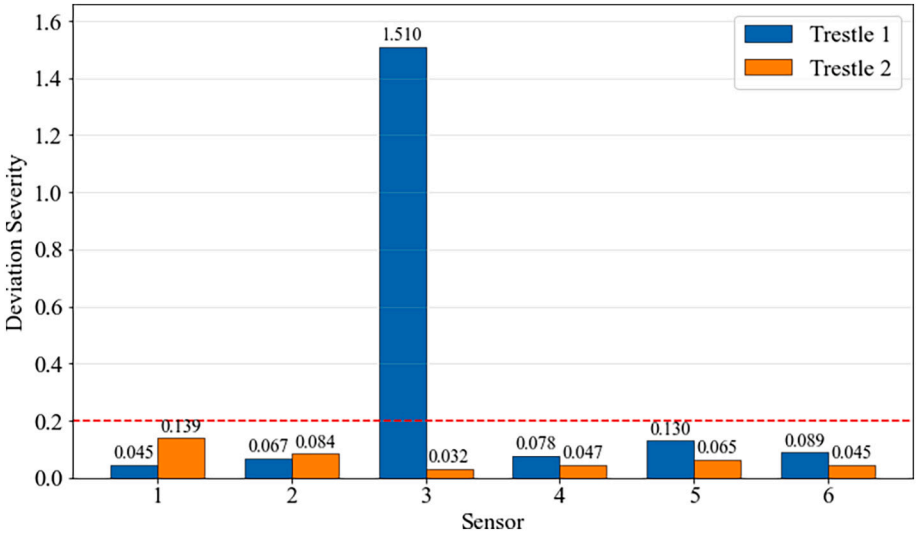
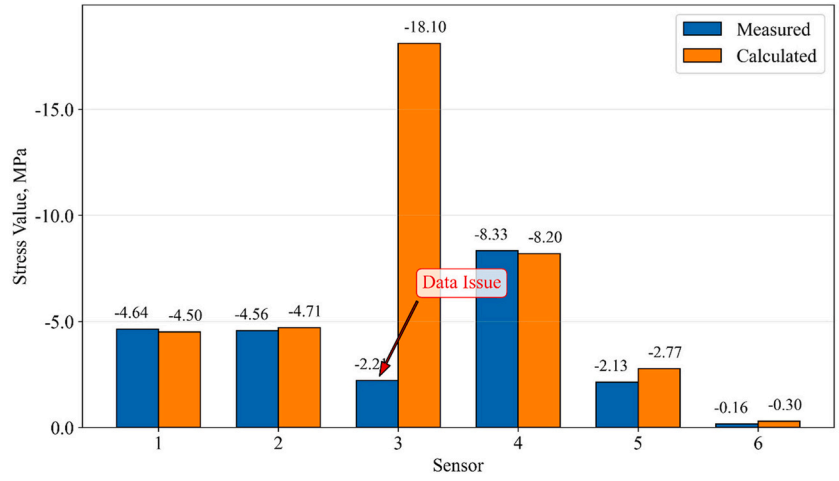
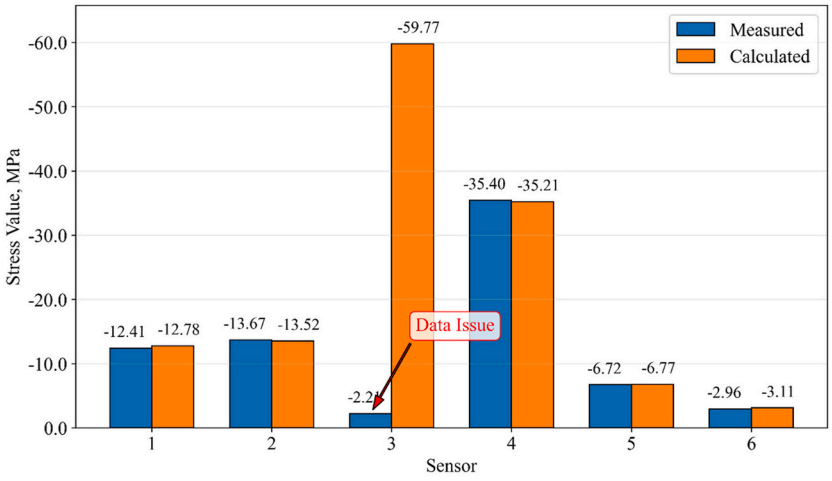


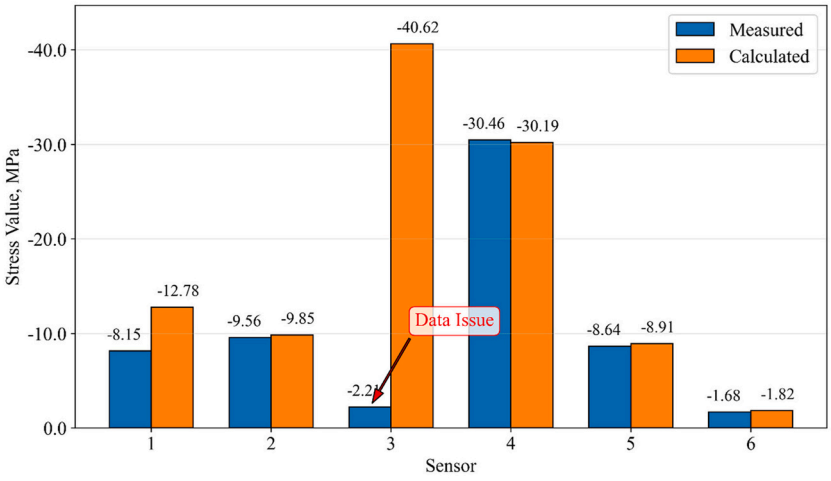
Fig. 9. Deviation severity of sensors.



(a) Scenario 1

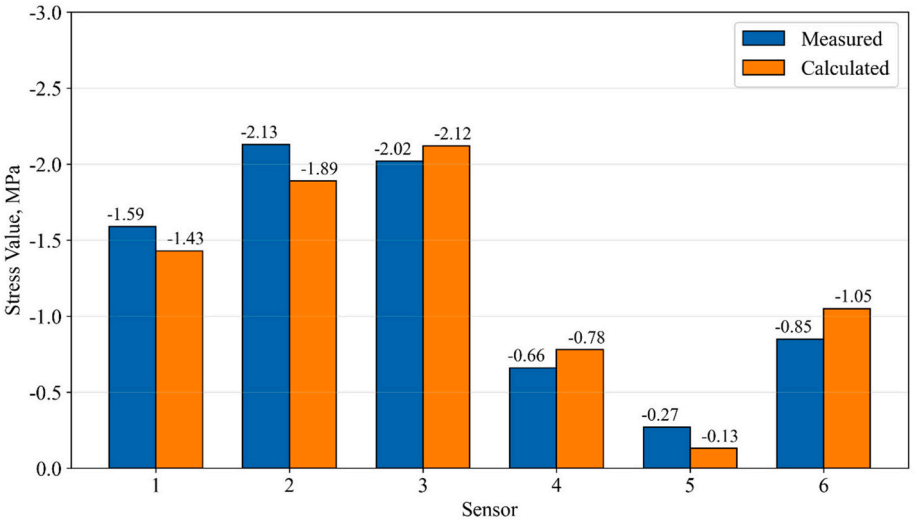


(b) Scenario 2

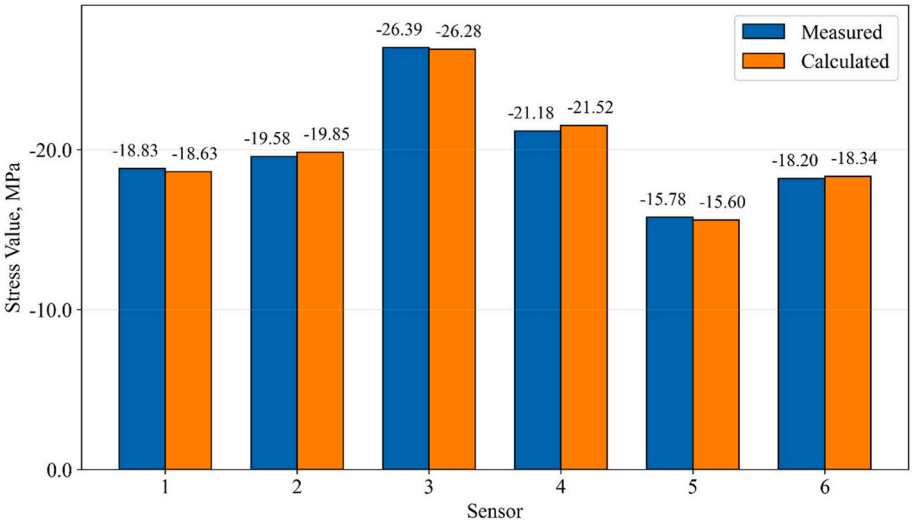


(c) Scenario 3

Fig. 10. Comparison of measured and calculated sensor stresses.



(d) Scenario 4



(e) Scenario 5

Fig. 10. (*continued*).

frequent objective function evaluations. To mitigate this limitation, data-driven surrogate models (e.g., Kriging, artificial neural networks), trained on representative FE simulation datasets, replace the full FE model during optimization [57–59]. Additionally, the standard PSO employs uniform behavioral patterns, update strategies, and weighting parameters across the swarm, potentially inducing premature convergence to local optima, insufficient balance between global exploration and local exploitation, and suboptimal convergence rates [60].

To mitigate these limitations, this study proposed a modified PSO (Fast-PSO) algorithm integrating dynamic population differentiation with adaptive group allocation. Furthermore, high-potential solution subspaces (HPSS) are dynamically identified and modeled using independent surrogate models, while direct mathematical optimization techniques are applied to accelerate solution convergence within these subspaces. As outlined in Algorithm 2 and Table 1, particles are classified into three behavioral groups, namely exploratory, exploitative, and stagnant, following multiple iterations of the standard PSO framework. This classification employs the Exploration Dynamism Index as Eq. 7 in Table 1, which quantifies convergence stagnation via the Euclidean distance between a particle's current and historical best positions. Exploratory particles exhibiting high dynamism retain standard PSO update rules to intensively probe unexplored regions of the solution space. Exploitative particles with moderate dynamism leverage low-fidelity surrogate models (e.g., quadratic response surfaces) to map localized HPSS, enabling rapid mathematical optimization techniques (e.g., gradient-based convergence) within these refined regions. Stagnant particles demonstrating minimal dynamism trigger Lévy-flight-aided relocation to escape local optima as Eq. 10 [61], redistributing them toward unexplored regions.

As formulated in Eq. 11, the algorithm monitors the evolution of the global best solution within the most recent five generations to determine whether the search process has stagnated, and adjusts the proportions of exploratory and exploitative particles accordingly. If no improvement is observed, the system identifies stagnation and increases the proportion (e.g., 5 %) of exploratory particles while reducing exploitative ones to enhance global search ability and escape local optima. Conversely, if improvement occurs, the system increases the proportion (e.g., 2 %) of exploitative particles to focus the search on promising regions and accelerate convergence. After the target proportions are set, particle classification in the current iteration is performed based on the

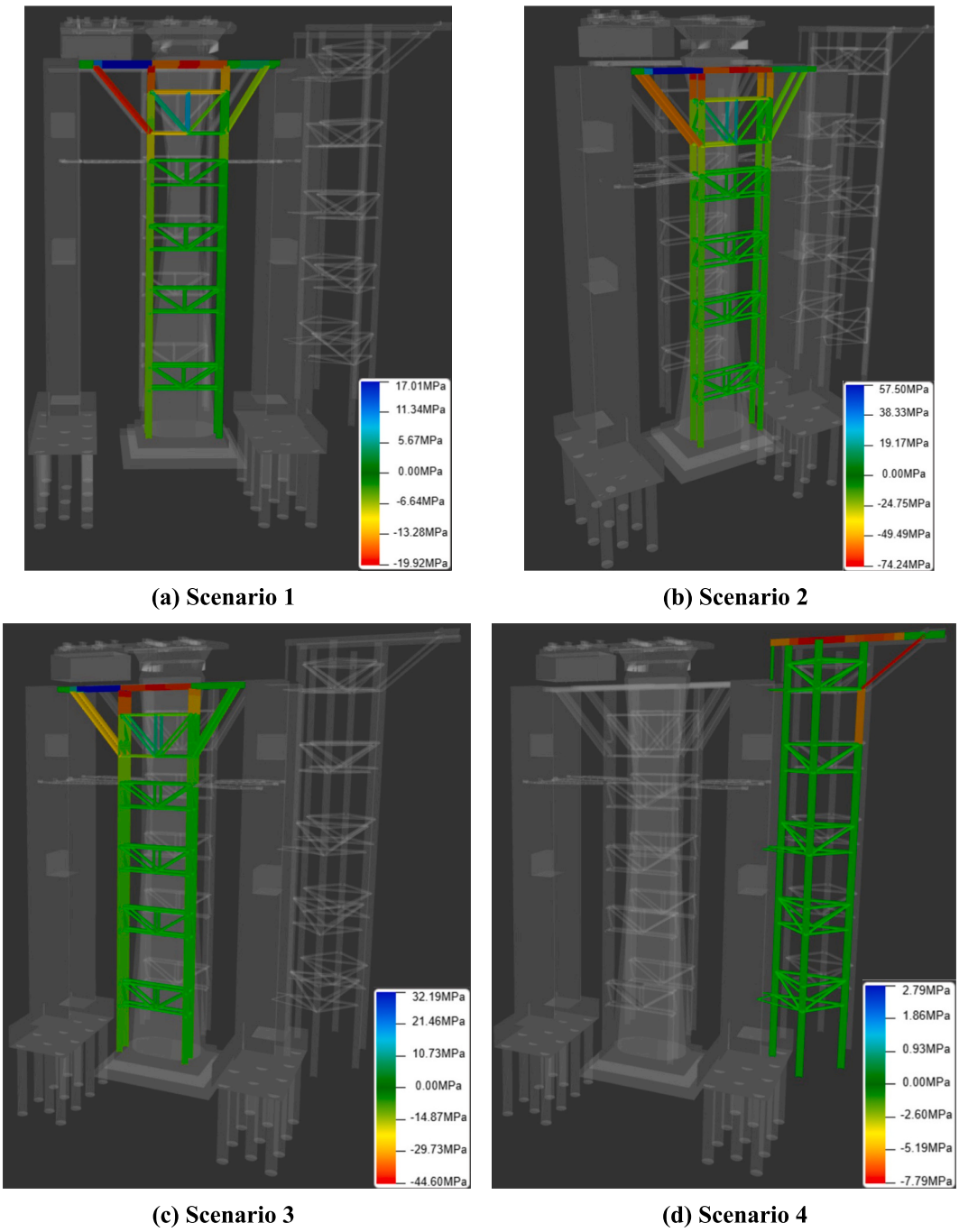
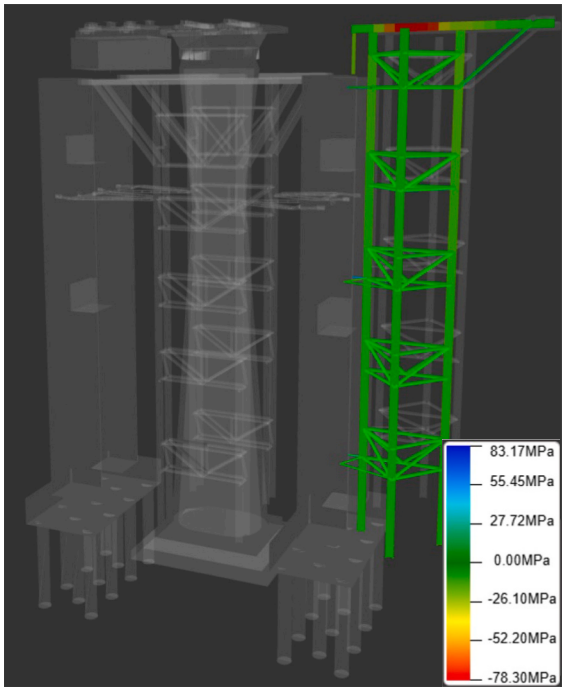


Fig. 11. Visualization of calculated global stress distribution.

movement range of each particle: those with the smallest ranges are assigned as stagnation-type; among the remaining, particles with the largest ranges are assigned as exploratory-type; and the rest are assigned as exploitative-type.

The proposed Fast-PSO systematically balances global exploration, local exploitation, and convergence recovery while preserving core PSO mechanics. Its innovations manifest in two primary mechanisms. First, dynamic particle categorization, guided by real-time convergence status and diversity feedback, enables efficient computational resource distribution. This allows strategic prioritization of critical search behaviors across optimization phases: emphasizing exploration for rapid identification of promising regions during initial stages, and shifting toward exploitation to accelerate convergence to precise solutions later. Such adaptive balancing mitigates redundant searches in low-potential regions and prevents premature convergence from insufficient exploration or excessive exploitation, thereby significantly reducing high-quality solution discovery time and directly enhancing optimization speed.

Second, the dynamically constructed surrogate models for HPSS leverage available real-evaluated data points to form local approximations using explicit and tractable mathematical representations. This formulation enables the direct application of efficient mathematical optimization techniques to rapidly obtain optimal solutions, thereby bypassing computationally expensive iterative particle evaluations. The resulting local optimum provides guidance for Exploitation particles, significantly accelerating convergence within potential basins and playing a decisive role in global optimization efficiency.



(e) Scenario 5

Fig. 11. (*continued*).

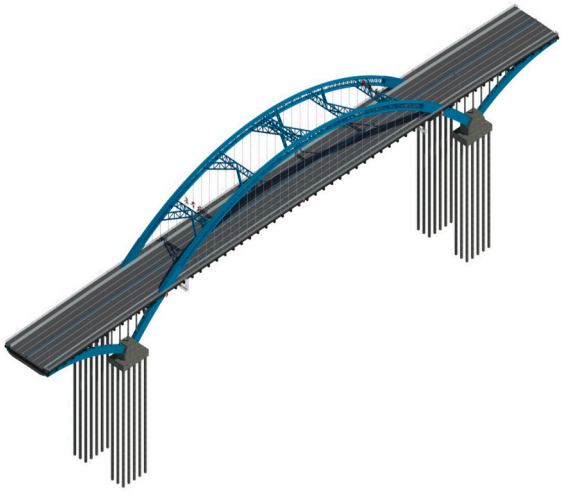


Fig. 12. Schematic diagram of the verification bridge model.

3. Engineering application of the proposed digital twin

This section introduces the engineering application used to demonstrate the implementation of the proposed DT framework. It presents the construction background, critical working scenarios, sensor deployment, and system configuration, forming the basis for the subsequent evaluation of the system's operational performance.

3.1. Background of the application

In the newly constructed Chengdu–Chongqing Middle Line, the Jingkou Double-Track Bridge intersects the existing Xinjingkou Jialing River Bridge on the Shanghai–Chengdu Line via a grade-separated crossing. As shown in Fig. 3, the two lines intersect at an angle of approximately 13°, with an overlap length of 115 m. To minimize the longitudinal elevation of the new line, an underpass scheme was adopted. The spatial overlap between the new and existing piers required the demolition of gravity piers No. 23 and No. 24 of the Xinjingkou Jialing River Bridge, which were replaced by portal piers to accommodate the new alignment. The entire construction process comprised over 20 tasks categorized into three phases: construction of the new portal piers, transfer of the existing bridge loads to these piers, and removal of the existing gravity piers.

This application focuses on five representative scenarios described in Table 2. As illustrated in Fig. 4, the steel crossbeam of the newly constructed portal pier rested on temporary steel tubular trestles (Trestle No. 1). The steel crossbeam bore the dead load transmitted from the main girder and underwent deformation. As a result, Trestle No. 1 was subjected to both the load transferred through the steel crossbeam and direct loading from the construction platform. The removal of existing pier caps required lateral displacement using winches mounted on sliding trestles (Trestle No. 2). As a result, Trestle No. 2 primarily resisted horizontal forces exerted by the winches while simultaneously supporting vertical loads from existing pier caps.

The construction process entailed significant complexity, tight scheduling, and elevated safety risks. Crucially, the trestles bore most structural loads, making their stability and safety paramount to project integrity. To ensure their structural integrity, stress sensors were installed at the most critical sections for real-time monitoring as illustrated in Fig. 5, which were identified through prior analysis. However, deviations from design conditions, including varying mechanical and human-induced loads and boundary condition discrepancies, limit the reliability of localized monitoring. It motivated the full-field stress distribution analysis to comprehensively assess structural behavior in this application.

3.2. Implementation of the digital twin system

Corresponding to the functional modules shown in Fig. 1, Fig. 6 illustrates the representative devices used in the proposed DT system,

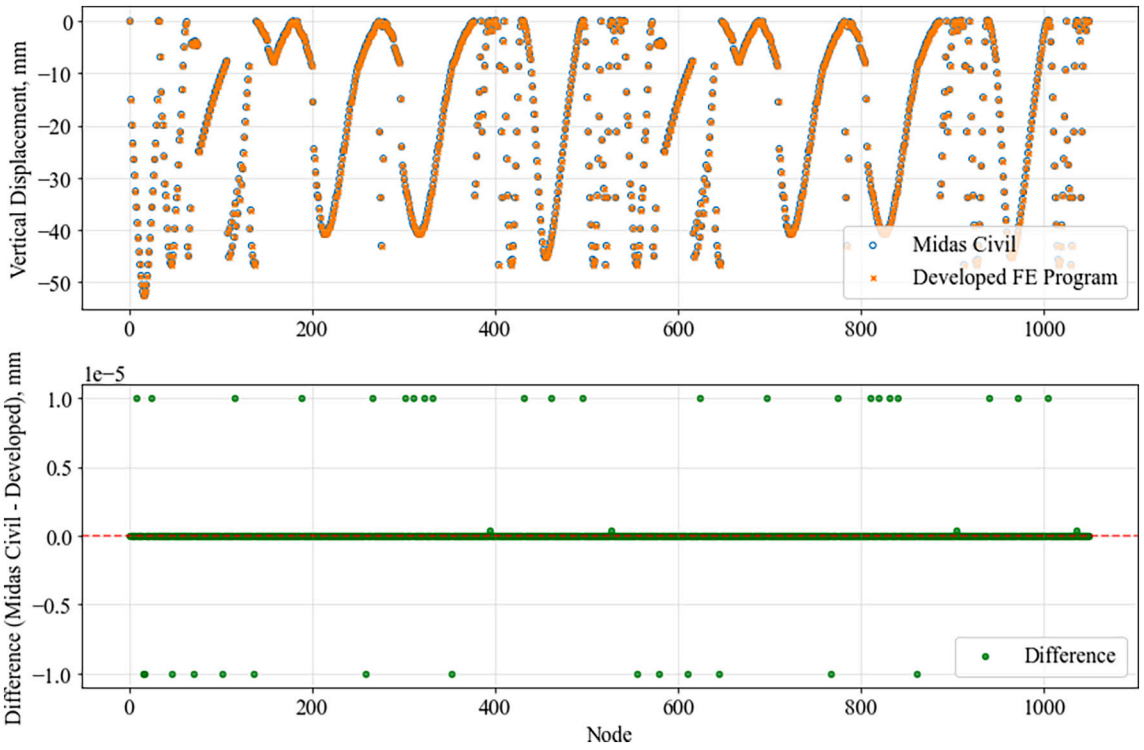


Fig. 13. Performance comparison of developed finite element program and Midas Civil.

with their configuration details provided in Table 3. The system employed a 5G mobile network for data transmission from the edge preprocessing layer to the cloud computing layer, taking advantage of its high bandwidth and low latency. Besides, it adopted an edge-cloud collaborative architecture in which the edge preprocessing layer performed lightweight local computations, including denoising, sampling, and data packaging. The reduced data volume to the cloud improved transmission efficiency and minimized network latency. In addition, the utilized edge device was equipped with a dedicated local storage unit of approximately 20 MB, capable of retaining about one day of sensor data. This capacity is sufficient to allow maintenance personnel to identify and rectify potential network interruptions in a timely manner. During periods of network disconnection, sensor data are temporarily buffered at the edge and automatically synchronized once connectivity is restored. Owing to the use of precise timestamps, this mechanism ensures data integrity and prevents any temporal inconsistency. Finally, online interaction was facilitated through a self-developed platform that enables interactive visualization, real-time monitoring, and efficient user–system communication throughout the DT workflow.

4. Results and discussion

This section comprehensively examines the performance of the proposed DT framework. First, the computational efficiency of the entire process, from data acquisition to stress distribution visualization, is quantitatively analyzed, with particular emphasis on evaluating the contribution of the two proposed strategies to improving overall efficiency. Subsequently, the effectiveness of the proposed damaged sensor identification strategy is demonstrated. Finally, by comparing the stress data collected from the physical entity with those predicted by the DT under five different scenarios, the accuracy and comprehensiveness of the twin's predictions, as well as its capability to identify potential structural risks, are validated.

4.1. Efficiency assessment

As shown in Table 4, the total time required from data acquisition to global stress visualization across all five scenarios does not exceed 0.3 s (300 ms), demonstrating the real-time capability of the DT system and its compliance with the fundamental requirements for risk early-warning tasks in SHM [62]. It is noteworthy that the time spent on hardware and network-related operations accounts for less than 20 % of the total duration. This observation underscores the significance and necessity of the efficiency enhancements introduced in this paper for the remaining modules, particularly structural simulation and system identification, which together account for the majority of the computational time.

The generation and assembly of the stiffness matrix constitute the establishment of the structural FE model, enabling the calculation of structural responses under applied external loads. Fig. 7 illustrates the specific impact of different stiffness matrix generation methods on the execution time for structural stress distribution calculation, in which the stiffness matrix (i.e., the FE model) is required to be invoked once. The results demonstrate that directly calling the precomputed stiffness matrix (requiring approximately 20–30 ms) reduced the execution time by up to 70 times compared to computing and calling the stiffness matrix (requiring approximately 1600–1900 ms). It can thus be anticipated that using precomputed stiffness matrices will lead to significant time savings in modules such as deviation severity determination of sensors and load identification, in which repeated invocation of the FE model is required. Given the assumptions adopted in this application, including material linearity, small deformations, and fixed boundary conditions, the use of this approach is well justified. For example, the temporary steel trestles examined here are expected to remain within the elastic range under service loads [63,64]. Moreover, Table 4 shows that the time required for global stress computation constitutes only 10–15 % of the total workflow duration, suggesting that the framework is likely to maintain satisfactory performance when applied to larger and more complex structures.

Fig. 8 illustrates the specific impact of different optimization

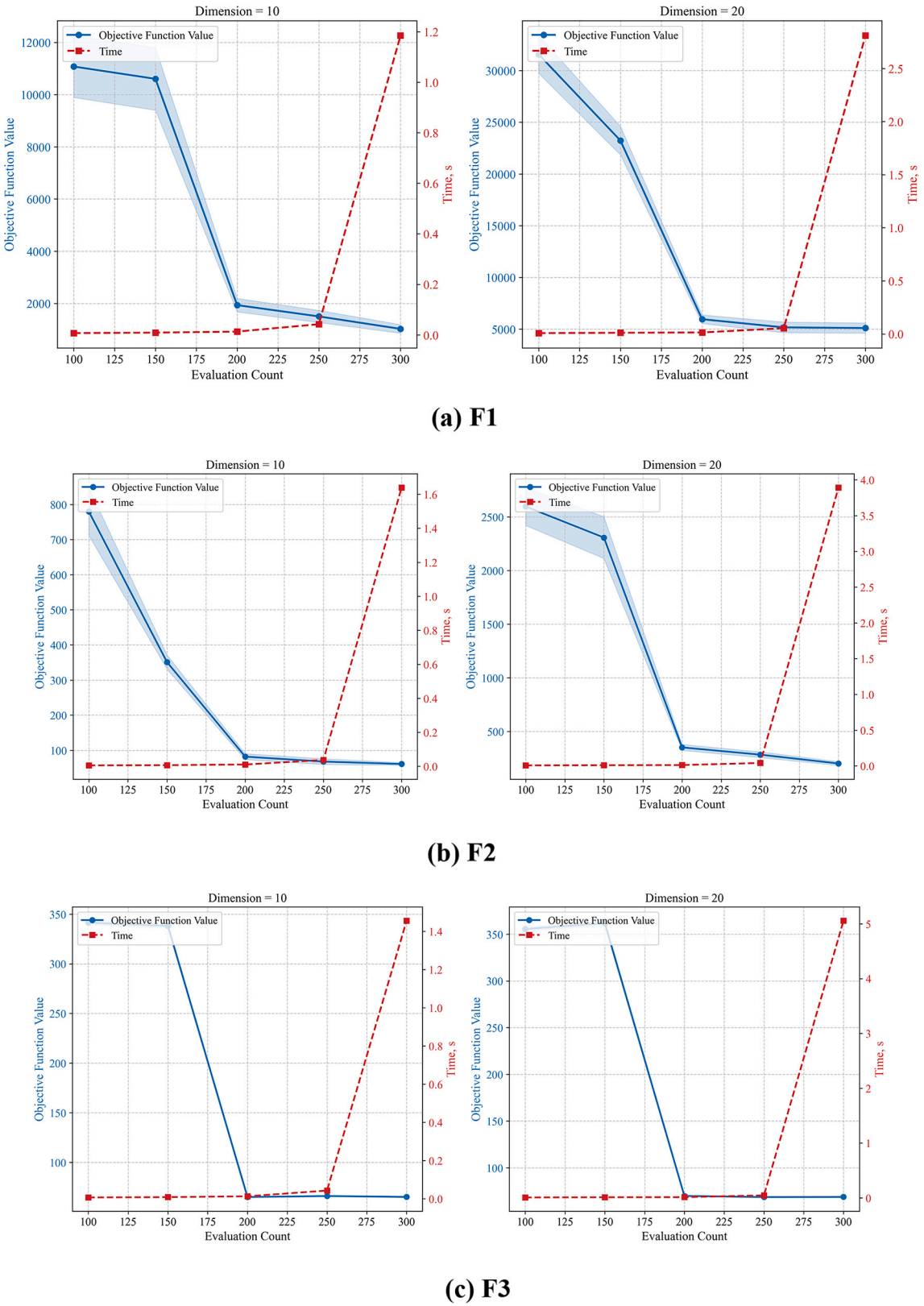
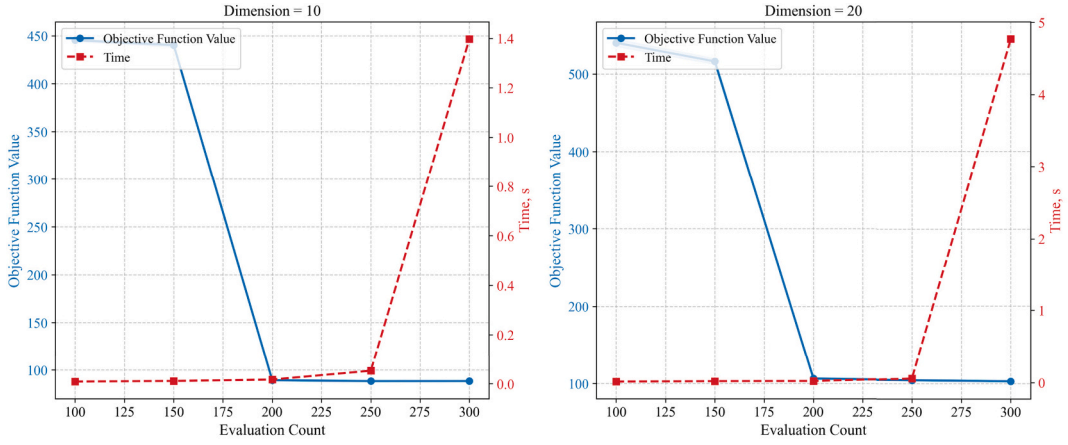
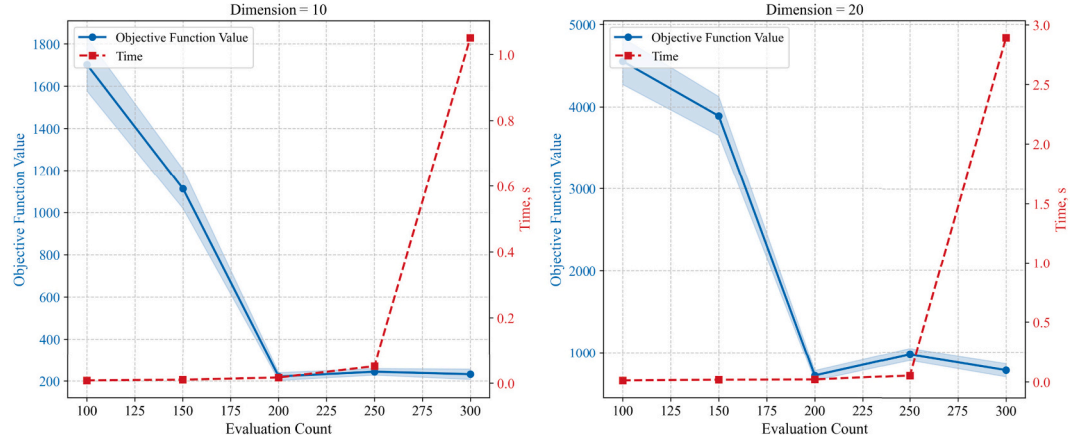


Fig. 14. Value and execution time of Fast-PSO versus objective function evaluation.

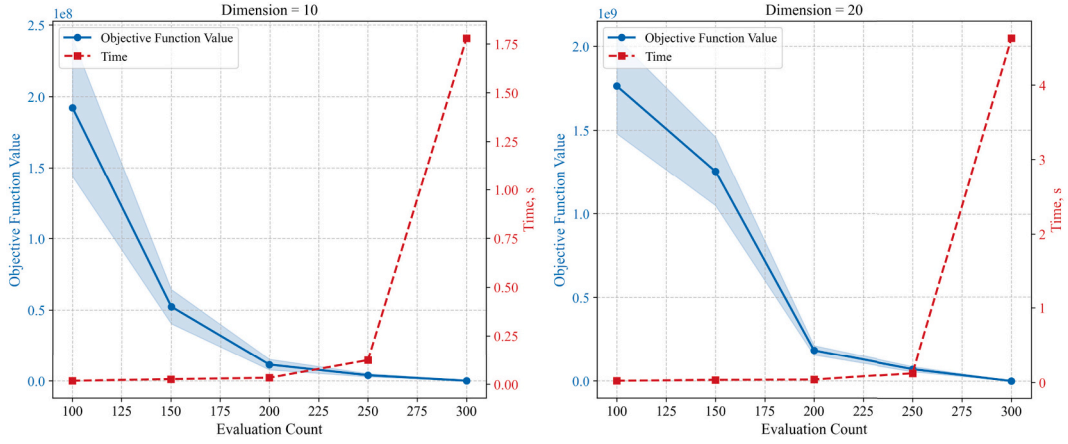
methods on the execution time for structural load identification. In this application, using the hyperparameters provided in Table 5 and achieving a convergent objective function value of 0.01, Fast-PSO reduced the computational time by approximately 70 % across all five scenarios compared to the standard PSO, decreasing from approximately 300–800 ms to 100–200 ms. These results demonstrate the successful coding and implementation of this modified optimization algorithm. Furthermore, as a general-purpose algorithm applicable to diverse



(d) F4



(e) F5



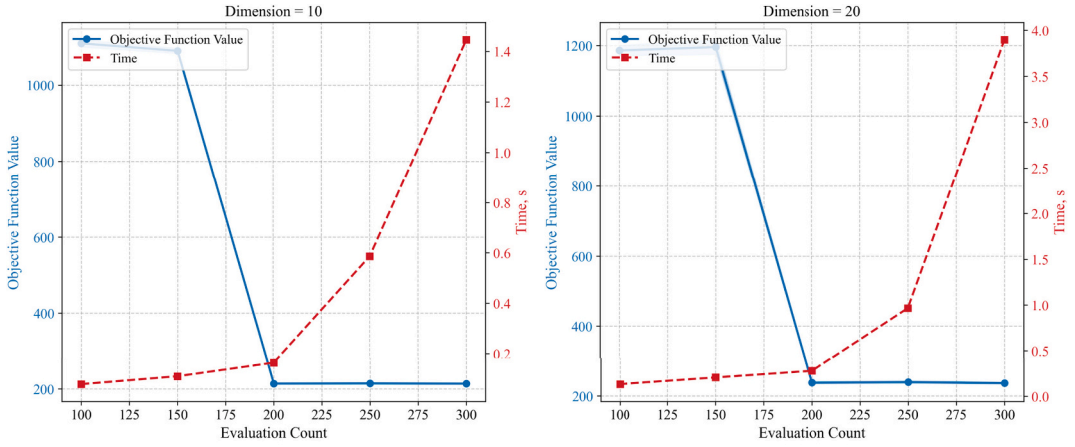
(f) F6

Fig. 14. (*continued*).

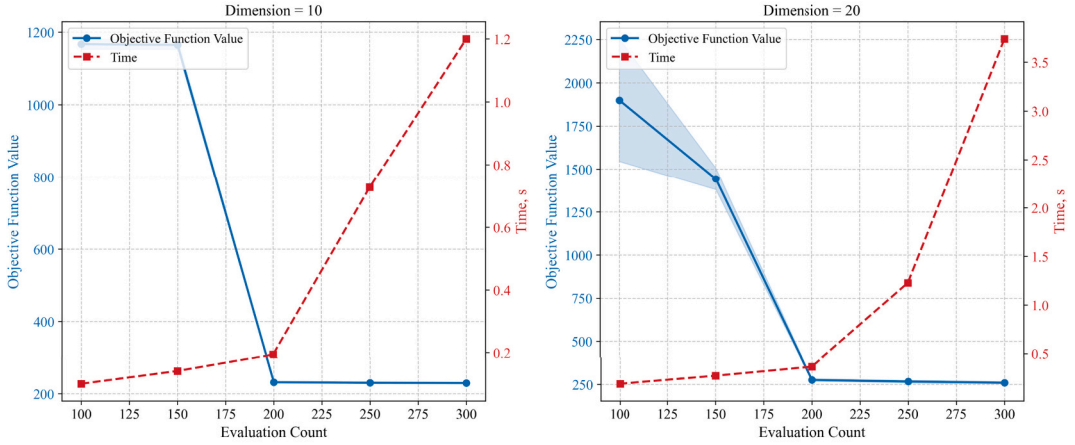
mathematical and engineering optimization problems, the superiority of Fast-PSO is more comprehensively demonstrated through comparative evaluations with the standard PSO on benchmark problem sets. The corresponding results are provided in the Appendix A.2 "Evaluation of Fast-PSO with benchmark functions".

4.2. Fidelity assessment

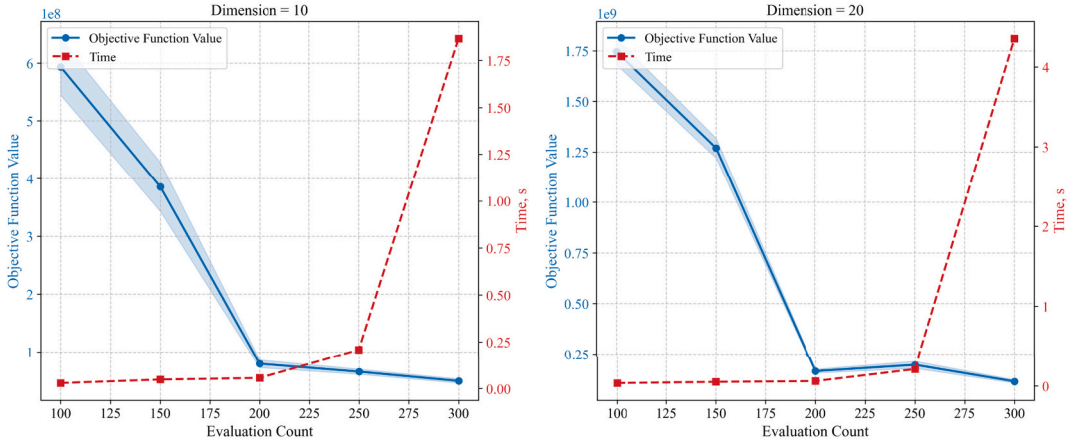
The deviation severity of all sensors calculated according to Eq. 4 is shown in Fig. 9. Among these, the deviation severity derived from data collected by Sensor 3 (installed on Trestle 1, corresponding to Scenarios 1–3) is significantly higher (exceeding 1.5) than those of the other sensors (less than 0.2). Therefore, during load identification using Fast-PSO, the weight of this sensor in the objective function was reduced (set



(h) F7



(i) F8

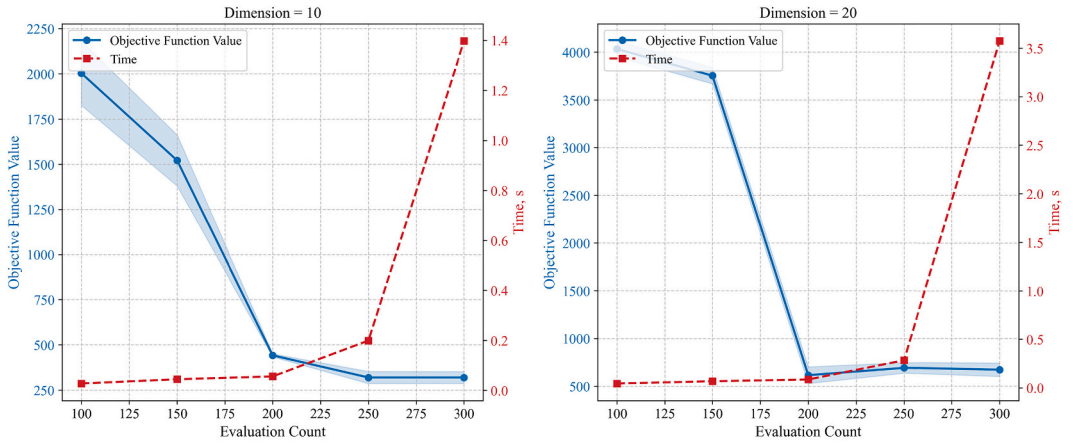


(j) F9

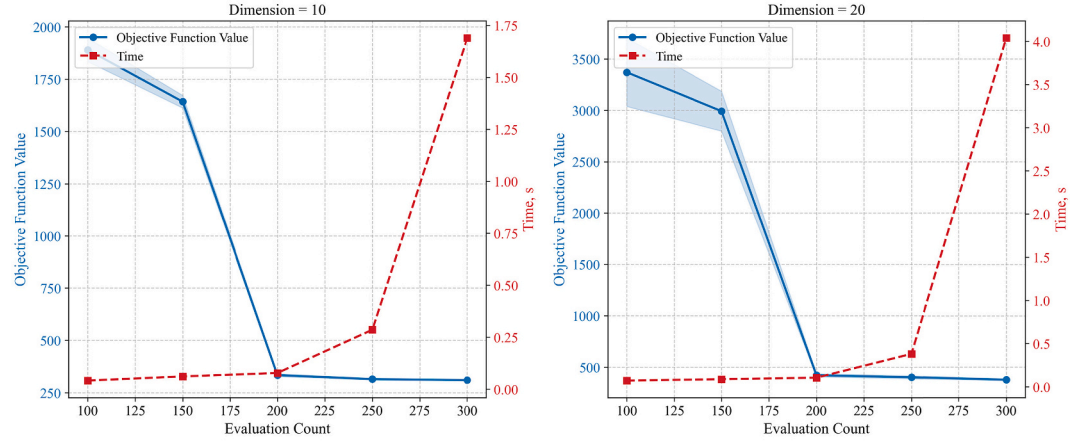
Fig. 14. (*continued*).

to an extremely small value). The accuracy of using deviation severity to identify sensor faults is corroborated by the stress measurements recorded in Fig. 10. It can be observed that the measurement from Sensor 3 under Scenarios 1–3 consistently register a value of −2.21 MPa (compressive stress), which clearly contradicts the actual mechanical behaviors of the structure under varying external loads.

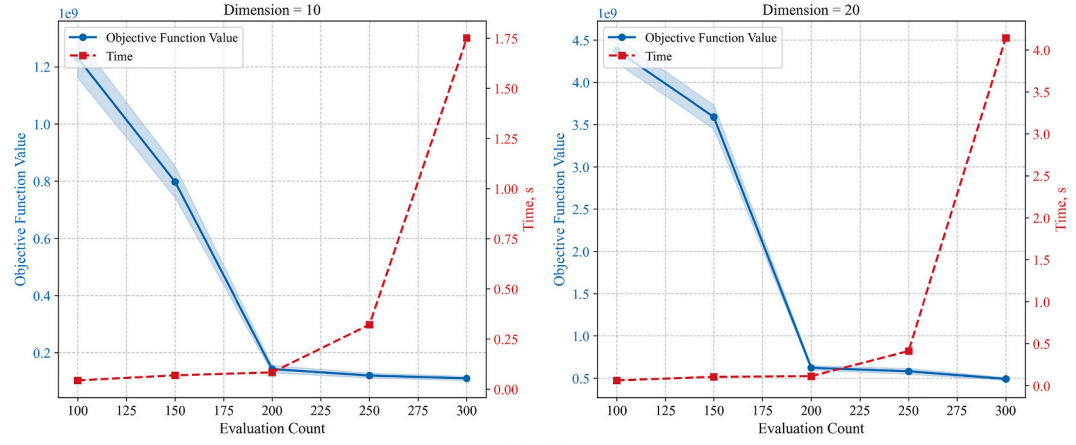
Furthermore, Fig. 10 presents a comparison between the stresses measured and calculated via the developed FE program with identified loads at the sensor locations. The results indicate a high level of agreement in most cases with relative errors less than 10 %, demonstrating the accuracy and reliability of the entire DT system.



(k) F10



(l) F11



(m) F12

Fig. 14. (*continued*).

4.3. Risk identification

Fig. 11 presents the global stress distribution of the structure calculated using the FE program with the identified loads, covering both sensor-instrumented and non-instrumented locations. In addition to clearly illustrating the varying stress levels across structural components, the stress nephograms for each scenario reveal that the computed extreme stress values exceed those measured from sensor-instrumented locations. For instance, as shown in Fig. 11(e), the maximum computed stress in Trestle 2 under Scenario 5 is −78.30 MPa, whereas the maximum measured stress at sensor locations shown in Fig. 10(e) is only −26.39 MPa. This discrepancy highlights a significant misalignment between the critical regions predicted in prior analyses and caused by the actual loads, which may be attributed to the differences between design assumptions and real-world conditions as previously discussed. These findings underscore the safety imperative of deploying a real-time DT system for structural state monitoring during construction, which constitutes the fundamental motivation of this study.

Table 6
Comparative performance of Fast-PSO and PSO under similar execution time.

Function	D = 10		D = 20		Function	D = 10		D = 20	
	PSO	Fast-PSO	PSO	Fast-PSO		PSO	Fast-PSO	PSO	Fast-PSO
	Mean	Mean	Mean	Mean		Mean	Mean	Mean	Mean
	STD	STD	STD	STD		STD	STD	STD	STD
	Best	Best	Best	Best		Best	Best	Best	Best
	Time, s*	Time, s	Time, s	Time, s		Time, s	Time, s	Time, s	Time, s
F1	**9.09E+02**	1.00E+03	1.31E+04	**5.27E+03**	F7	2.06E+03	**2.15E+02**	2.20E+03	**2.36E+02**
	6.06E+02	**3.48E+02**	6.31E+03	**1.19E+03**		1.50E+01	**3.04E+00**	6.69E+01	**1.02E+01**
	4.04E+02	4.87E+02	6.56E+03	**3.23E+03**		2.03E+03	**2.06E+02**	2.11E+03	**2.21E+02**
	0.097	0.094	0.114	0.105		1.503	1.152	2.547	2.267
F2	4.45E+02	**6.74E+01**	5.82E+02	**2.74E+02**	F8	2.25E+03	**2.30E+02**	2.31E+03	**2.56E+02**
	3.79E+01	**1.20E+01**	**7.91E+01**	8.81E+01		2.40E+01	**5.15E+00**	7.31E+01	**1.50E+01**
	4.08E+02	**4.71E+01**	4.61E+02	**1.36E+02**		2.23E+03	**2.25E+02**	2.23E+03	**2.32E+02**
	0.081	0.076	0.097	0.088		1.972	1.529	3.449	3.096
F3	6.17E+02	**6.41E+01**	6.46E+02	**7.00E+01**	F9	3.61E+08	**6.10E+07**	8.85E+08	**1.45E+08**
	7.83E+00	**7.29E-01**	1.37E+01	**1.77E+00**		7.28E+07	**8.48E+06**	2.37E+08	**2.69E+07**
	6.06E+02	**6.25E+01**	6.26E+02	**6.71E+01**		2.06E+08	**4.45E+07**	5.16E+08	**8.79E+07**
	0.095	0.083	0.120	0.102		0.528	0.463	0.560	0.519
F4	8.48E+02	**8.73E+01**	9.60E+02	**1.04E+02**	F10	2.71E+03	**3.39E+02**	4.87E+03	**6.41E+02**
	1.11E+01	**9.38E-01**	1.87E+01	**1.91E+00**		2.94E+02	**9.35E+01**	1.49E+03	**2.02E+02**
	8.25E+02	**8.50E+01**	9.28E+02	**9.94E+01**		2.50E+03	**2.51E+02**	2.51E+03	**2.61E+02**
	0.131	0.107	0.134	0.136		0.518	0.431	0.778	0.679
F5	1.03E+03	**2.29E+02**	3.95E+03	**8.37E+02**	F11	3.05E+03	**3.13E+02**	3.56E+03	**3.88E+02**
	1.27E+02	**4.93E+01**	1.44E+03	**1.70E+02**		3.40E+01	**5.57E+00**	1.72E+02	**1.59E+01**
	9.04E+02	**1.57E+02**	1.59E+03	**5.63E+02**		3.01E+03	**3.07E+02**	3.30E+03	**3.54E+02**
	0.133	0.106	0.134	0.126		0.810	0.601	1.053	0.911
F6	**2.77E+04**	5.57E+06	**7.01E+06**	1.09E+08	F12	9.10E+08	**1.15E+08**	3.96E+09	**5.56E+08**
	3.87E+04	4.20E+06	**8.80E+06**	4.95E+07		4.35E+07	**1.59E+07**	3.44E+08	**1.00E+08**
	6.52E+03	7.43E+05	**4.44E+05**	2.29E+07		8.79E+08	**9.48E+07**	3.53E+09	**4.35E+08**
	0.291	0.241	0.309	0.287		0.826	0.636	1.117	0.955

* Benchmark function evaluations were performed with 256 GB of memory, with all other settings identical to those in the engineering application.

5. Conclusions

This paper presented an efficiency-enhanced digital twin (DT) system tailored to the real-time requirements of structural health monitoring. A bridge pier replacement project was selected as an engineering application, where six pre-installed sensors were used to identify external forces and compute stress distributions of two steel trestles under five representative scenarios. The main conclusions are as follows:

- The DT system adopted edge-cloud integration, cloud computing, and 5G mobile network at the hardware level, and employed a Python-based finite element (FE) program and swarm-intelligence-based optimization at the software level. The total time from data acquisition to stress visualization remained below 300 ms, demonstrating strong real-time capability and suitability for risk warning in health monitoring.
- The developed FE program, combined with precomputed and fixed stiffness matrix, significantly accelerated load identification and global stress computation, reducing FE computational time to 1/70 of the original.
- The proposed deviation severity coefficient with adaptive weighting accurately identified abnormal sensors, ensuring prediction-measurement stress errors below 10 % in most cases.
- The modified particle swarm optimization (Fast-PSO) algorithm, incorporating population differentiation and local acceleration, improved both accuracy and efficiency. It reduced computational time by approximately 70 % in the application and achieved up to a tenfold accuracy improvement over the standard PSO on 12 CEC'2022 benchmark problems.
- The DT system effectively identified critical stress regions, with predicted maximum stresses consistently exceeding those considered during design, in some cases by more than threefold.

Overall, this paper presented an effective solution for real-time monitoring of engineering structures and offered technical guidance for efficient simulation and system identification. While some proposed approaches were developed under certain simplifying assumptions, such as linear elasticity and small deformations, these do not diminish their practical relevance for most engineering applications. Future research will aim to further enhance the framework's adaptability and computational robustness, enabling its extension to more complex scenarios, including systems with varying stiffness or nonlinear responses.

CRediT authorship contribution statement

Yazhou Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yong Deng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Gongjian Zhou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Jiwei Zhong:** Writing – review & editing, Supervision, Investigation. **Xungang Zhao:** Writing – review & editing, Investigation, Data curation. **Chenxu Zhou:** Writing – review & editing, Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Validation of the finite element implementation and Fast-PSO performance

This appendix provides supplementary validation for the methods presented in this paper. The first part verifies the developed finite element program, while the second evaluates the performance of the proposed Fast-PSO algorithm.

A.1. Validation of the developed finite element program

This section validates the accuracy of the developed Python-based FE program. The verification case involves a half-through concrete-filled steel tubular (CFST) tied-arch bridge. As shown in Fig. 12, the bridge has a total length of 1130 m, with a main span of 252 m, side spans of 60 m, and a width of 27 m. All structural components were assigned a density of 7.85 g/cm^3 and an elastic modulus of 200 GPa.

The bridge model was constructed within the self-developed Python program following a standard FE modeling procedure, whose general framework is described in Section 2.2 "Python-based finite element program with cloud computing". The bridge geometry was first drafted in Autodesk CAD based on the design drawings. The Python program then extracted the three-dimensional coordinates of key feature points to generate primary nodes. Additional nodes were created by linear interpolation along the geometric centerlines of the main components (e.g., girders, arch ribs, and hangers). These continuous components were discretized into a series of connected three-dimensional beam elements. To ensure computational accuracy, denser mesh divisions were applied in regions with expected high stress gradients, such as near joints and supports. The final FE model consisted of 1050 nodes and 6300 degrees of freedom.

Fixed boundary constraints were applied at both bridge supports, and the self-weight of the structure was considered as the applied load. Before computation, the program automatically executed the load application process by traversing each beam element, calculating its volume and weight based on sectional properties and material density, and converting the weight into an equivalent set of nodal forces and moments acting at both ends of the element. Finally, all equivalent nodal loads were systematically assembled into the global load vector for subsequent structural response analysis.

The vertical nodal displacements under self-weight load were computed and compared with the results obtained from the commercial FE software Midas Civil. The results presented in Fig. 13 demonstrate close agreement between the two methods, confirming the accuracy of the developed program. Furthermore, the computational times were 2.09 s for Midas Civil and 1.85 s for the developed program, reflecting the earlier conclusion that by retaining only core modeling and solving modules, the present FE program achieved higher computational efficiency compared to commercial software. This advantage makes it a suitable tool for real-time SHM applications.

A.2. Evaluation of Fast-PSO with benchmark functions

This section conducts a comprehensive performance evaluation and comparison between Fast-PSO and the standard PSO using the CEC'2022 test suite, which consists of 12 rotated, shifted, hybrid, and composite minimization functions [65]. The aim is to demonstrate the computational efficiency improvements of Fast-PSO in a more standardized manner through widely recognized benchmark problems.

Table 5 presents the hyperparameter settings used for both the engineering application and the benchmark problems. No hyperparameter optimization was performed; instead, classical parameter values commonly adopted in previous studies were directly employed [66]. The primary objective is to validate the performance improvements introduced by the Fast-PSO strategies, such as population differentiation and surrogate-model acceleration, over the standard PSO. Therefore, to ensure a fair and scientifically rigorous comparison, the principle of controlled variables was applied: the baseline parameters of Fast-PSO and the standard PSO were kept identical. This design ensures that any observed performance gains can be attributed to the proposed mechanisms rather than from any incidental advantage due to hyperparameter tuning. For the benchmark problems, both algorithms were evaluated and compared under two relatively high dimensions (D = 10 and D = 20) in the following two aspects:

- Fast-PSO was repeated 30 times for each of five different numbers of objective function evaluations (100, 150, 200, 250, and 300). The mean and confidence interval of the final objective function values, as well as the average execution time, were recorded. The results are presented in Fig. 14.
- Both Fast-PSO and the standard PSO were repeated 30 times under approximately the same execution time (with 270 objective function evaluations for Fast-PSO and 2000 for the standard PSO). The mean, standard deviation, and best values of the final objective function values were obtained. The results are summarized in Table 6.

The results in Fig. 14 demonstrate that the objective function value decreased as the number of function evaluations increased, indicating a correct optimization trend of the modified algorithm. The narrow confidence intervals further reflect the high stability of its optimization performance. Due to strategies such as population differentiation which introduce higher complexity and more sophisticated algorithmic behavior, the per-evaluation time of Fast-PSO is longer than that of the standard PSO. However, Fast-PSO explores the solution domain more extensively and facilitates more efficient information exchange among particles within a single evaluation process. As a result, it achieved better solutions within the same total execution time.

As shown in Table 6, across all 12 benchmark functions and two dimensions, Fast-PSO outperformed the standard PSO in nearly all cases in terms of the mean, standard deviation, and best values of the objective function. Specifically, the accuracy metric (i.e., the mean objective function value) shows an improvement of up to an order of magnitude.

Data availability

The data that support the findings of this study are within the article, and the code of Fast-PSO applied in this study is provided on GitHub at https://github.com/pakanama/Fast-PSO.

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